



System Architecture Primer

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This document explains why a system architecture is important to planning and operation of technology infrastructure. A published architecture for enterprise systems is an important unifying force and will assist service units to invest appropriately.

What is a System Architecture?

A *system* is defined as “a group of interrelated elements forming a collective entity.” An *architecture* is “a style and method of design and construction.” Together, these describe a method for creating cooperating groups of elements. We will define a *system architecture* as

“A set of agreed-to methods, standards, and techniques for building complex functions from simpler components that promotes efficient division of responsibility, enables incremental change, and increases functional reliability.”

System architectures are common, although they may not be referred to as such. The evidence of system architectures can include industry standards, governmental regulation, and commodity products. The following could serve as examples:

- Electric power system. Standards for electrical outlets, voltages, distribution, and hierarchical “grids” describe a conscious architecture for sharing and using electricity. System architectures can vary from country to country.
- Fire protection. Municipal codes for building construction, building access, location of water sources, pumping stations, and firehouses reflect a systematic approach to protection of life and property.
- Facsimile. Fax machines are commodity products that interoperate without significant effort from the purchaser. Scanning and encoding techniques, inter-machine negotiation of speed and resolution, and fax software for computers define, or benefit from, a system architecture.

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Why are System Architectures Important?

System architectures simplify the management of complex environments. Most importantly, system architectures partition the delivery of the end product into discrete internal services that pass work from one to another to accomplish the goal. To be judged most successful, this service-oriented internal partitioning will be invisible to the consumer of the end product, projecting the appearance of an integrated whole.

As an example, consider a television set. This device relies upon two systems: the electrical power system and the broadcast television system. In the case of the electrical system, the consumer inserts the plug into a corresponding standard receptacle for that voltage. The craftsman who installed the receptacle did not have to be an expert in the entire electric power infrastructure of the United States, but only in the portion from the local distribution to the home, through the fuse panel and to the outlet. While possibly ignorant of any aspect of the electric wiring, the consumer is confident in the service because of building codes (standards) that must be met for the house to be declared safe and construction completed.

Because a system architecture partitions the infrastructure, changes to the overall service can be made more easily. As the sub-components pass work between them, each views the others as “black boxes” with only inputs and outputs. If the same results can be achieved in new ways, then the sub-component can be improved in isolation from the balance of the system.

Returning to the television set example, the broadcast system is one which has undergone many changes with individual sub-components improving one after another. Figure 1 shows a simplified system architecture for television service. The live event is captured with a camera and passed to a programming service which could store the event or pass it immediately to the broadcast service. The broadcast service distributes the signal to the display device (the television set) where the consumer realizes the goal of the service.

Figure 2 shows steps in the evolution of television service. From its 1940's beginnings in black-and-white services through color to cable distribution to new programming models such as pay-per-view, to high-definition television, each step has been an evolution in one or more portions of the architecture. Market forces make these changes gradual – not discontinuous. If one broadcaster switched to only HDTV broadcasts, the initial number of viewers would be very small since the installed base of display devices would lag the change.

In summary, a system architecture defines sub-component services that pass work between them and can be constructed individually, managed separately, and changed in isolation - provided that the standardized services between sub-components are preserved.

How Do System Architectures Improve Organizational Effectiveness?

For a business or university, system architectures offer the greatest flexibility for managing the delivered service. By understanding the system architecture, and especially the sub-components, administrators can determine which portions are strategic and which are less important. This understanding influences investment of money, human resources, and management time.

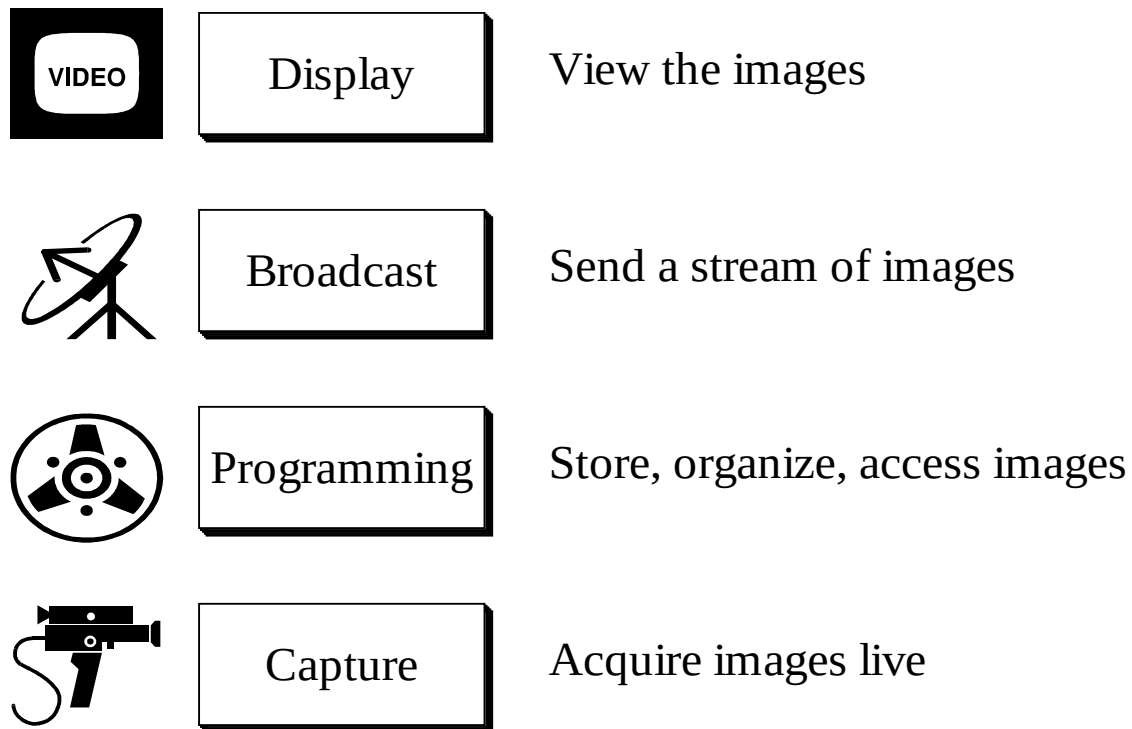


Figure 1 – Television System Architecture

Television Migration

Incremental changes in the system architecture enable new services with minimal disruption.

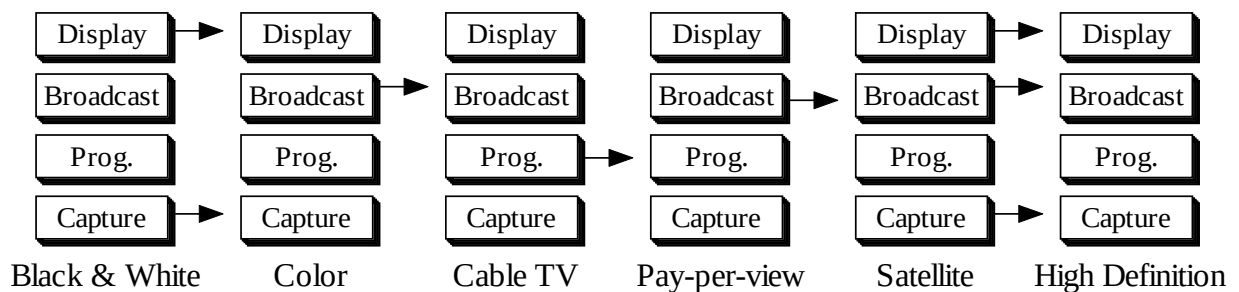


Figure 2 - Changes to Television System Architecture Over Time

The processes for purchasing goods make up a system architecture that involves the purchaser, a purchasing department staff, a financial system, and the vendor, supported by communication services (see Figure 3). In a traditional paper-based system, the purchaser makes a vendor and

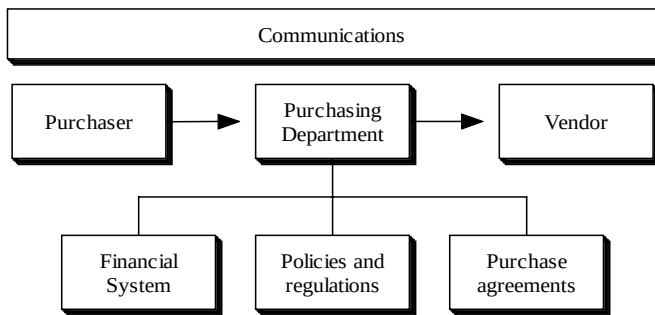


Figure 3 – Purchasing System Architecture

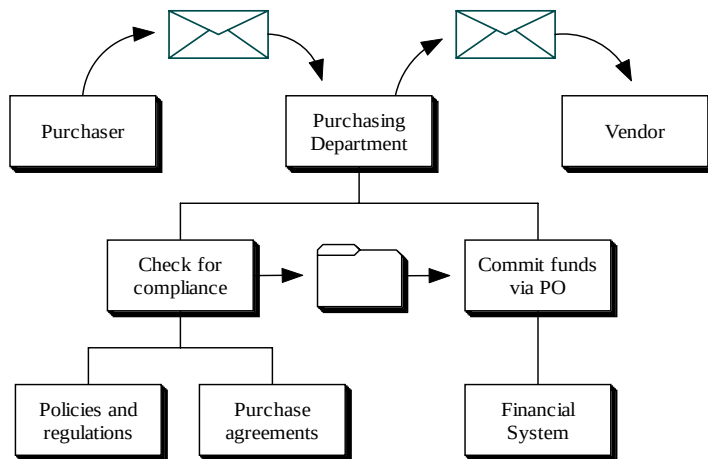


Figure 4 - Services within the Architecture

product decision and mails it to the purchasing department. That department checks for sufficient funds, regulatory compliance, access to volume discount programs, etc., prepares a purchase order within the financial system and mails it to the vendor. In this scenario, the service-oriented approach between sub-components can be defined more precisely (Figure 4) and examined for possible changes.

With evolutionary changes in the communication methods (e.g. from mail to corporate networks) new opportunities emerge. In the eyes of the purchaser, the service of the purchasing department is to accept the request, qualify the request against a set of rules and then use the financial system to issue the purchase order to the vendor. In a classic change to “self-service”, this sub-component could be changed to be a central computer-based service to accept electronic purchase requests, apply the same rules, and direct the financial system to create the purchase orders for mailing. With appropriate inter-company

technologies, the purchase order could be transmitted to the vendor immediately. See Figure 5.

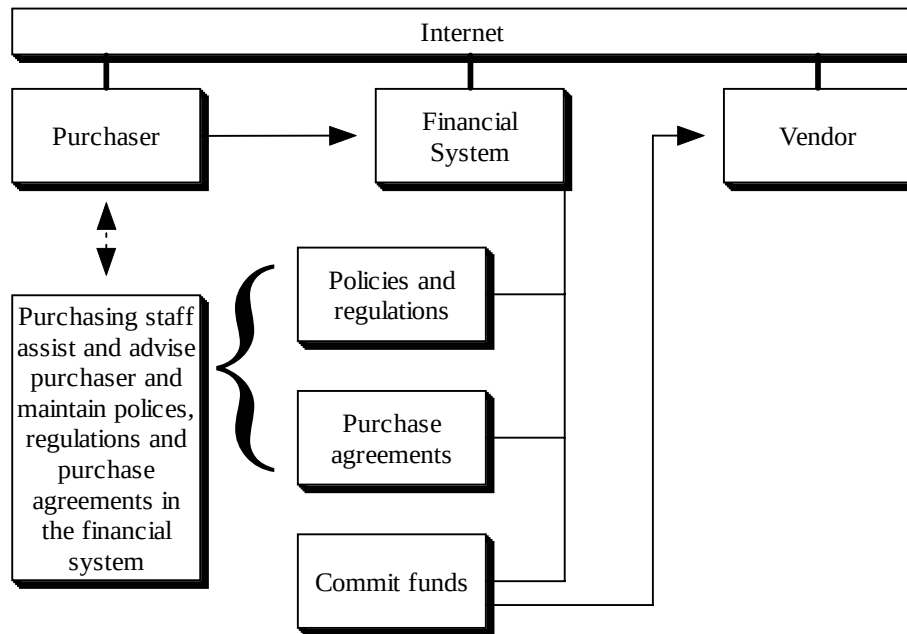


Figure 5 – A Modified Architecture for Self-Service Purchasing

What is the benefit of these changes? First, time is saved. Handling of the request is eliminated. Lag time between request and notification of the vendor can also be cut dramatically. Second, purchasing staff resources can be directed to new beneficial services, such as assisting the initial purchase decision-making or negotiating better contracts for large purchase volumes. Third, the expectations of the purchaser are set more precisely. With the lack of internal handling, a purchase order is known to be at the vendor and not within the internal business processes.

How Will an Enterprise System Architecture be Manifest at Northwestern?

The system architecture of the greatest current interest is that which must be defined to encompass the enterprise systems of the University. In this context, “enterprise systems” include administrative systems and major academic systems such as course management, library, and school or division applications with requirements for central administrative data.

A simplified conceptual diagram (Figure 6) divides the services of enterprise systems into sub-components representing the user, the communication network, core data, core services, and higher services. Improving administration of the University implies that services to the user (faculty, staff or student) will be improved in speed, convenience and accuracy. It also requires securing data and services; accommodating best-of-breed core services; and enabling electronic commercial partnerships. These improvements will require not only modification of the existing sub-components, but the creation of new services and standards to be incorporated between sub-components.

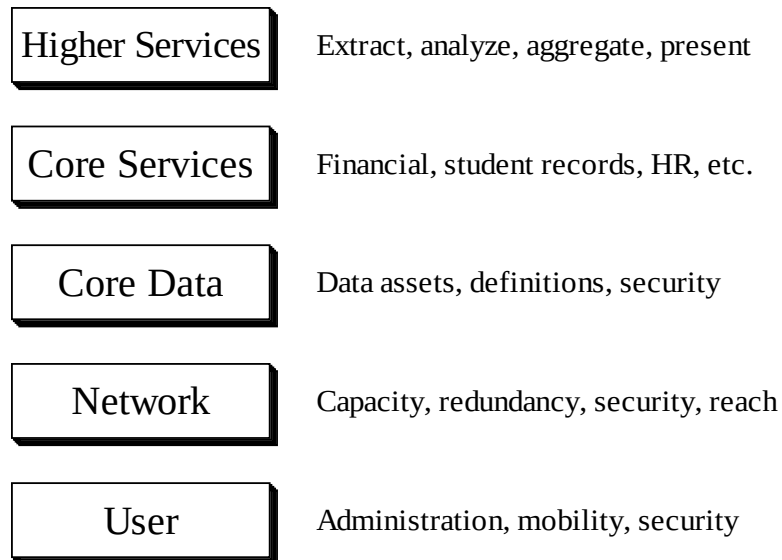


Figure 6 - Simplified Enterprise System Architecture

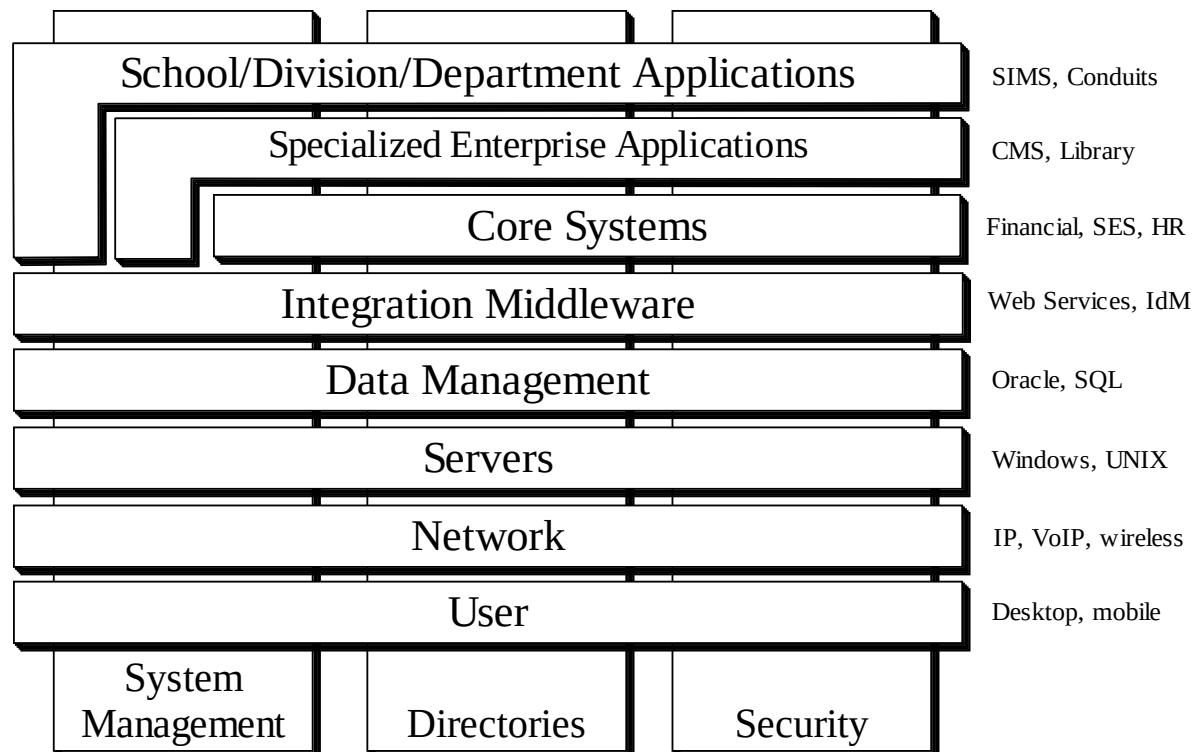


Figure 7 – A More Complete System Architecture for Northwestern

The visible result of these changes will be central management of key services. For example, the creation and management of electronic identities for access to University resources must be a closely managed service. Confidence and trust in electronic identities is essential in order to deploy necessary access controls and permissions. Another example is inter-application communications that will result in choreographed services across administrative systems based upon single events (such as employee hiring or termination). Without a standard and stated direction for this communication technique, separate administrative systems will remain isolated. The additional detail implied by these additional services is reflected in a more complete architectural model in Figure 7.

Summary

A publicized and endorsed system architecture improves many aspects of a service environment:

- Sub-components can be changed as technology improves without adverse effects on the ultimate service delivered
- A “service-oriented” approach to the sub-components opens possibilities for outsourcing or elimination of unnecessary components.
- Maintenance and management of sub-components can remain separated and follow best-of-breed trends.
- Extensions and improvements to the service can be rapidly tested and deployed due to published sub-component functions.

Northwestern’s administrative systems must be embedded within an overall system architecture to focus future efforts, enhance security, and ultimately deliver the best services possible.